

COURSE NUMBER: MA1905

COURSE TITLE: Problem Solving for IT

COURSE DESCRIPTION:

The course guides students in building logical thinking and problem-solving skills essential for computer programming in the context of Information Technology. This course offers a practical foundation in mathematics relevant to business data processing, encompassing basic algebra, calculus, and various computer-related mathematical concepts. By the end of this course, students will have acquired the critical skills and knowledge necessary to excel in Information Technology and its numerous real-world challenges.

PREREQUISITES: None

CO-REQUISITES: None

CARNEGIE UNIT: Three (3)

CREDIT VALUE: Four (4)

COURSE HOURS PER WEEK: Four (4)

LAB HOURS PER WEEK: Zero (0)

REQUIRED TEXT:

Basta, A., DeLong, S., & Basta, N. (2014). *Mathematics for information technology*. Boston, MA: Cengage Learning. e-Text ISBN: 9798214348582

LEARNING RESOURCES: None

MAJOR TOPICS:

- 1.0 Basic Algebra
- 2.0 Problem-Solving Concepts
- 3.0 Numeration Systems
- 4.0 Program Structure
- 5.0 Logic
- 6.0 Boolean Algebra
- 7.0 Set Theory

LEARNING OBJECTIVES:

1.0 Basic Algebra

- 1.1 Define the terms:
 - 1.1.1 Constants
 - 1.1.2 Variables
 - 1.1.3 Equations
 - 1.1.4 Functions
- 1.2 Express values using exponents and scientific notations
- 1.3 State the order of operations for mathematical calculations
- 1.4 Solve problems requiring the calculation of percentages
- 1.5 Solve problems on a calculator using mathematical calculations for:
 - 1.5.1 Real numbers
 - 1.5.2 Exponents
 - 1.5.3 Percentages
- 1.6 Solve problems using inequalities
- 1.7 Solve problems requiring the transposition of equations

2.0 Problem-Solving Concepts

- 2.1 Define problem-solving
- 2.2 Discuss types of problems
- 2.3 Differentiate between problem-solving with computers and general problem solving
- 2.4 Distinguish between constants and variables
- 2.5 Describe data types
- 2.6 Develop expressions and equations

3.0 Numeration Systems

- 3.1 Describe a decimal number in expanded notation
- 3.2 Express a decimal number in expanded notation
- 3.3 Describe a binary number in expanded notation
- 3.4 Express a binary number in expanded notation
- 3.5 Describe an octal number in expanded notation
- 3.6 Express an octal number in expanded notation
- 3.7 Describe a hexadecimal number in expanded notation
- 3.8 Express a hexadecimal number in expanded notation
- 3.9 Convert whole numbers among the following systems:
 - 3.9.1 Decimal
 - 3.9.2 Binary
 - 3.9.3 Octal
 - 3.9.4 Hexadecimal

- 3.10 Use binary numbers to do the following:
 - 3.10.1 Add
 - 3.10.2 Subtract
 - 3.10.3 Multiply
- 3.11 Using hexadecimal numbers:
 - 3.11.1 Add
 - 3.11.2 Subtract
 - 3.11.3 Multiply
- 3.12 Use the AND operator to compare two binary values

4.0 Program Structure

- 4.1 Discuss how to organize problems
- 4.2 Analyze problems
- 4.3 Develop logic for problem solutions using a logic tool
- 4.4 Write an algorithm that solves a problem
- 4.5 Explain the purpose of documentation in the creation of a problem solution

5.0 Logic

- 5.1 Discuss with examples:
 - 5.1.1 Simple statement
 - 5.1.2 Compound statement
 - 5.1.3 Conditional statement
 - 5.1.4 Bi-conditional statement
 - 5.1.5 Inverse statement
 - 5.1.6 Negation
- 5.2 Construct truth tables for logic statements
- 5.3 Write algorithms using conditional logic to solve a problem

6.0 Boolean Algebra

- 6.1 Define Boolean algebra
- 6.2 Explain the Boolean operations of addition
- 6.3 Explain the Boolean operations of multiplication
- 6.4 Explain Boolean algebra properties
- 6.5 Explain the properties of Boolean algebra for networks
- 6.6 Simplify Boolean expressions
- 6.7 Write algorithms using Boolean algebra to solve a problem

7.0 Set Theory

- 7.1 Define a set
- 7.2 Discuss the meaning of symbols in standard set notation
- 7.3 Discuss the properties of sets using standard set notation
- 7.4 Apply the properties of the following elements:
 - 7.4.1 One-to-one correspondence
 - 7.4.2 Universal sets
 - 7.4.3 Equal sets
 - 7.4.4 Equivalent sets
 - 7.4.5 Subsets to sets
- 7.5 Discuss the standard set operation notations of:
 - 7.5.1 Union
 - 7.5.2 Intersection
 - 7.5.3 Complement
- 7.6 Use set operations to solve problems including:
 - 7.6.1 Performing unions
 - 7.6.2 Performing intersections
 - 7.6.3 Performing complements
- 7.7 Solve problems using sets
- 7.8 Define relations and functions using set-theoretic notation

EVALUATION:

Tests:	30%
Assignments:	35%
Final Exam:	35%

DATE DEVELOPED: October 2023

DATE REVIEWED:

REVISION NUMBER: 1

DATE REVISED: May 2025

PeopleSoft Effective Date: September 2025

Note to instructor: Check PIRS to ensure this outline is the most current version.